25901240 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Core Course (DSC) - MG4DSCECC200 - ELECTRONICS SERVICE TECHNOLOGY

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

MCQ

Answer all questions

Each question carries **1** marks

1. If cable insulation is intact but conductor temperature is high, the fault is: [U] / [CO1]

   a). Short circuit b). Overload

   c). Earth fault d). Open circuit

2. Thermistors are generally made of: [U] / [CO2]

   a). Pure metals b). Metallic alloys

   c). Semiconductor materials d). Insulating materials

3. When testing an NTC thermistor, its resistance decreases when: [A] / [CO3]

   a). Temperature decreases b). Temperature remains constant

   c). Temperature increases d). Voltage increases

4. Which material can be cut using a jigsaw? [U] / [CO2]

a). Wood only b). Metal only

c). Plastic only d). Wood, plastic, and thin metal

5. Which defect can be visually identified during PCB inspection? [U] / [CO3]

a). Incorrect resistance value b). Broken track

c). Frequency drift d). Signal delay

6. Which instrument measures electrical power directly? [U] / [CO1]

a). Ammeter b). Voltmeter

c). Wattmeter d). Energy meter

7. Why are angle gauges preferred over sine bars for small angular settings? [K] / [CO2]

a). Require less skill b). Faster setup and high accuracy

c). Lower cost d). No surface plate needed

8. Incorrect identification of transistor pins may cause: [K] / [CO3]

a). Improved performance b). No effect

c). Circuit malfunction d). Reduced power loss

9. Which fault causes fuse to blow frequently? [K] / [CO1]

a). Open circuit b). Short circuit

c). Loose connection d). Normal load

10. The working principle of a traditional AC clamp meter is based on: [U] / [CO2]

a). Hall effect b). Electromagnetic induction

c). Electrostatic force d). Thermoelectric effect

[1x10 = 10]

Part B

Short Answer Questions

Answer any **4** questions

Each question carries **5** marks

11. What is voltage? Explain its physical significance. [K] / [CO1]

12. Why is taper measurement important in mechanical and electronics equipment assembly? [U] / [CO2]

13. Differentiate between single-sided and double-sided PCBs. [U] / [CO3]

14. What is an electromagnetic flow meter? State its basic application. [K] / [CO2]

15. Explain how a wiring diagram helps in servicing electronic equipment. [A] / [CO3]

16. What problems may occur if wiring color codes are not followed? [K] / [CO1]

[5x4 = 20]

Part C

Essay Questions

Answer any **2** questions

Each question carries **10** marks

17. Define an ultrasonic flow meter. Explain its basic principle of operation. [K] / [CO2]

18. Define load requirement calculation. Explain why it is necessary in electrical installations. [U] / [CO1]

19. Explain how a BJT can be tested for proper functioning using a multimeter. [U] / [CO3]

20. Explain the importance of RCCB in electronics servicing. Compare RCCB with MCB. [U] / [CO1]

[10x2 = 20]